

Two Sides of the Same Coin: How LLMs Reveal Dual Narratives in Annual Reports

A novel semantic scoring framework for quantifying distinct risk and response narratives within corporate financial filings (Form 10-K).

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The Challenge: Nuance in Financial Narratives

10-K Reports are Strategically Designed

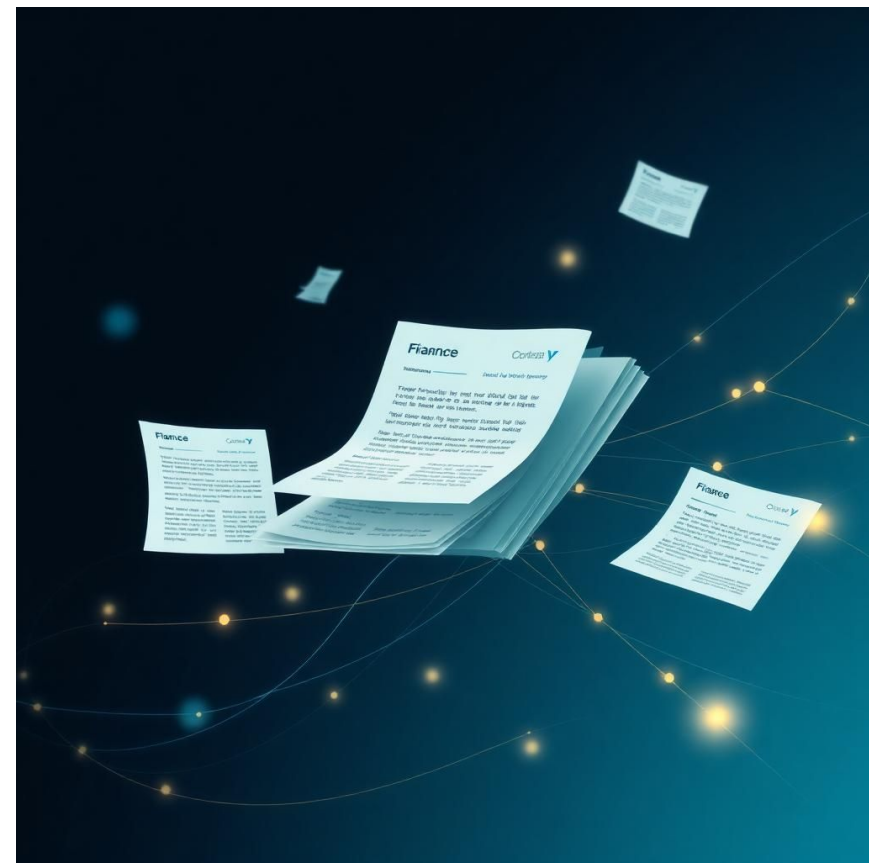
Beyond compliance, they serve as a platform for impression management and signaling future trajectory to stakeholders.

Traditional NLP Falls Short

Existing methods (dictionary-based, readability) capture superficial linguistic cues, missing deep, context-dependent financial interpretation.

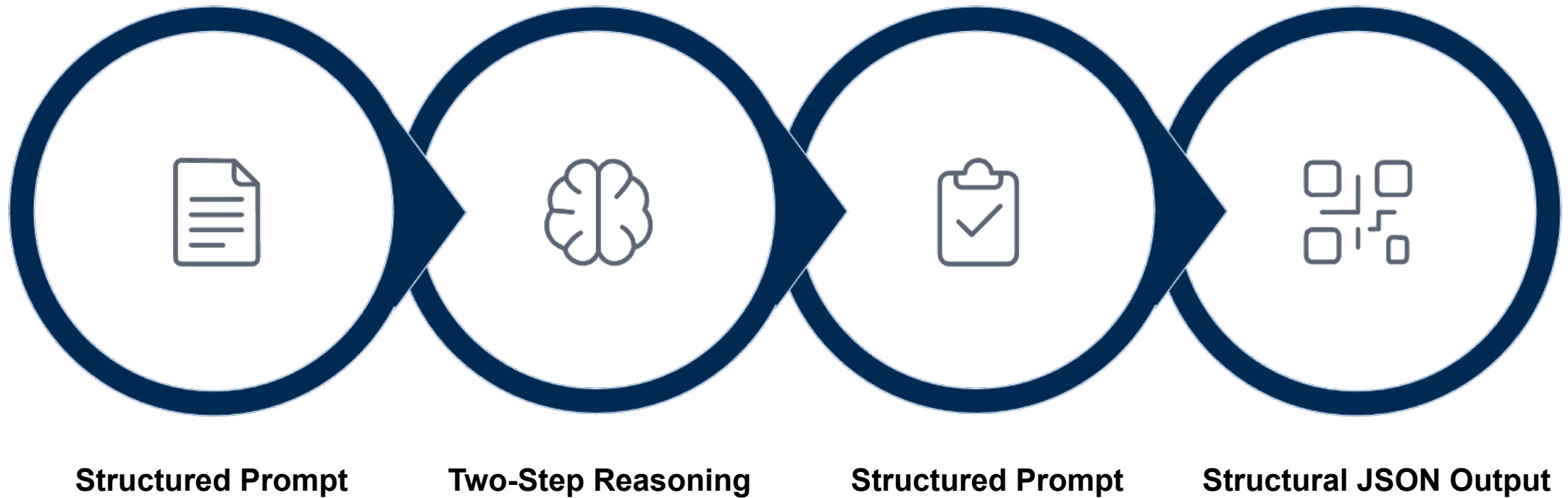
LLMs Lack Transparency

Advanced LLM comprehension is powerful, but internal mechanisms are opaque, hindering scrutiny and verification in high-stakes financial analysis.



Dual Narrative Analysis (DNA) Framework

We introduce a Chain-of-Thought (CoT) driven methodology to transform opaque LLM analysis into a transparent, verifiable semantic scoring engine.



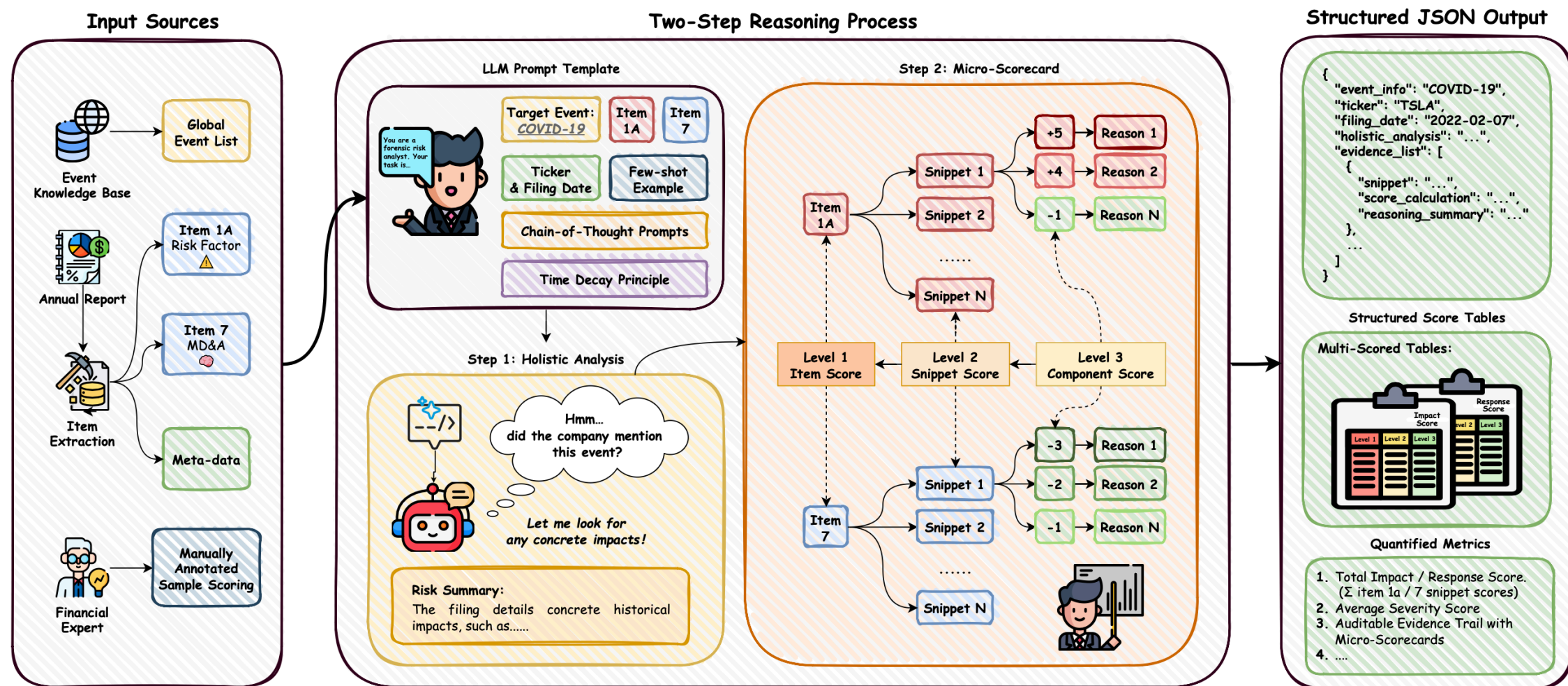


Figure 1: DNA Framework Using Chain-of-Thought (CoT) Reasoning for Quantifying Narratives on Global Events in Annual Reports (Form 10-K).

Dual Narrative Analysis (DNA) Framework

Our hierarchical scoring process ensures every score is tied to explicit textual evidence, ensuring verifiability.

Level 1: Item Category

LLM determines the corresponding 10-K section (Item 1A or Item 7).

Level 3: Component Scoring

Most granular level: LLM assigns positive (aggravating) or negative (mitigating) points to specific semantic components.

Level 2: Snippet Identification

LLM isolates text snippets relevant to the target global event (e.g., COVID-19).

Bottom-Up Aggregation

Component scores aggregate to Snippet scores, retaining detailed reason chains for audit requirements.

Quantifying the Dual Narrative

The DNA framework separates risk perception from managerial action, analyzing two critical sections of the 10-K filing.



Item 1A: Risks Factors

Focus: Legally mandated disclosure of material risks.

- ◆ Formal, legalistic, conservative tone.
- ◆ Quantifies the **Impact Score** (severity of potential negative consequences).



Item 7: MD&A

Focus: Management's strategic perspective and narrative freedom.

- ◆ Proactive, explanatory, forward-looking tone.
- ◆ Quantifies the **Response Score** (proactivity and quality of management's countermeasures).

Quantifying the Dual Narrative

The DNA framework separates risk perception from managerial action, analyzing two critical sections of the 10-K filing.

Example: Risk Impact Scoring

Snippet:

*"The Company's business, results of operations and financial condition, as well as the price of the Company's stock, have been **adversely affected** and could in the future be **materially adversely affected** by the COVID-19 pandemic."*

Score Components:

- **[+5]** → States that COVID-19 has already hurt the business (historical, direct impact).
- **[+2]** → Adds that future impacts could be 'materially adverse', signalling severity.
- **[-1]** → No quantitative magnitude provided.

→ **Final Snippet Score** = **[+5]** + **[+2]** + **[-1]** = **[+6]**

Reasoning Summary:

🌀 LLM → *"Direct admission of past and potential material harm warrants a high score even without quantification."*

📄 AAPL | 2021-10-29 | Item 1A: Risk Factors

Example: Risk Response Scoring

Snippet:

*"PCA **did not experience significant disruptions** in its operations as a result of the pandemic and has **maintained adequate availability of its workforce and supply of raw materials and services** to continue to serve its customers."*

Score Components:

- **[-3]** → Affirms absence of significant operational disruptions (mitigating).
- **[-1]** → Indicates stable workforce and supply chains (additional mitigation).

→ **Final Snippet Score** = **[-3]** + **[-1]** = **[-4]**

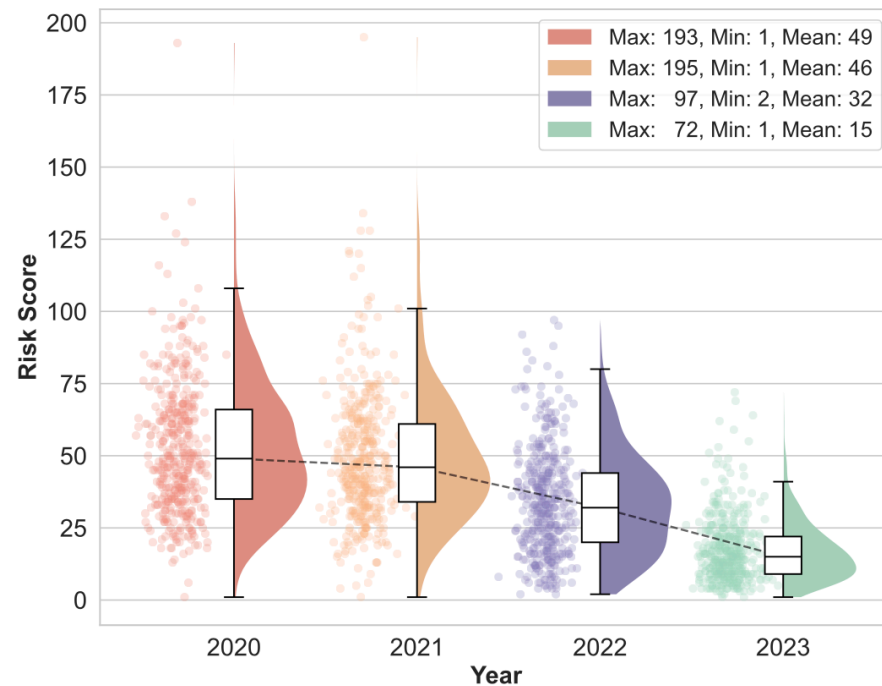
Reasoning Summary:

🌀 LLM → *"Purely mitigating language produces a negative risk score (-4), reflecting reduced severity."*

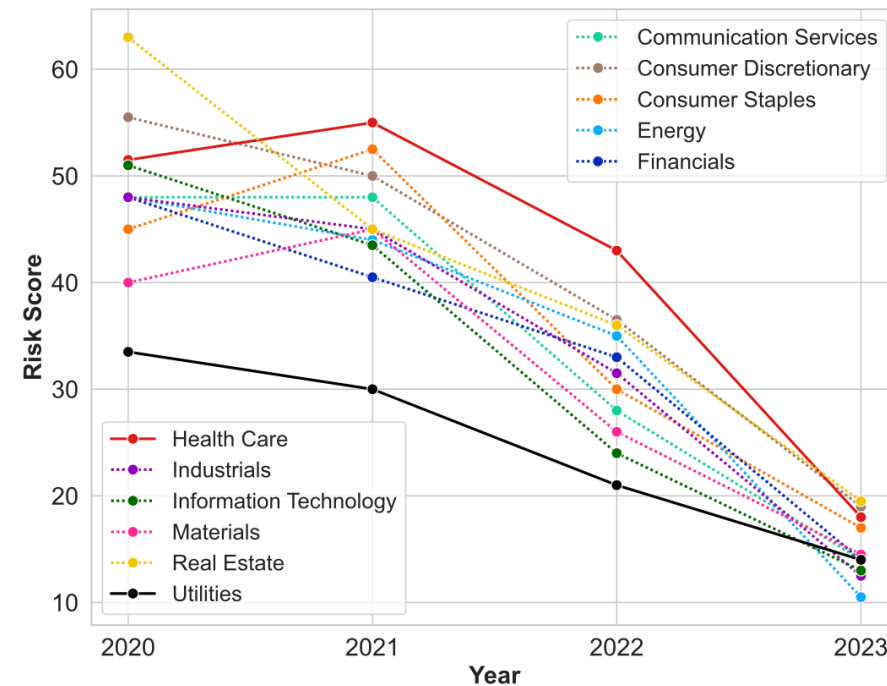
📄 PKG | 2021-02-24 | Item 7: MD&A

Case Study: COVID-19 Impact Score Trends

Analysis of Item 1A Impact Scores (2020-2023) reveals a normalization in perceived risk over time among S&P 500 firms.



(a)



(b)

Figure 3: Evolution of Firm-Level Impact Scores (2020 – 2023). (a) yearly score distributions across all S&P 500 firms. (b) yearly mean scores by GICS sector.

Response Score: Signal of Management Quality

The Response Score from Item 7 quantifies the intensity and quality of strategic managerial action.

Score Range	Response Level	Sample Score [†]	Example Snippet	Rationale / Interpretation
≤ -2.5	<i>Significant Mitigation</i>	-4	"We have ... enabled employees to work from home ...; adjusted work spaces and shift schedules ...; required face coverings and distributed personal protective equipment (PPE) ."	Extensive operational mitigation: covers remote work, on-site protocols, sanitation, and PPE—indicating a comprehensive COVID-19 response.
$(-2.5, 0)$	<i>Partial Mitigation</i>	-2	"In addition, when negative economic conditions have been particularly severe, like during the Covid-19 pandemic, we have moved decisively to lower operating expenses to improve financial performance."	Suggests a mitigating response through cost reduction, but the absence of concrete measures or operational details limits the strength of evidence—warranting a moderate mitigation score of -2 .
0	<i>Neutral / Indeterminate</i>	0	"We continued to see the commercial aerospace end market recover from the COVID-19 pandemic and were encouraged by the progress of the commercial aerospace market recovery."	Not actively mitigated, but partially mitigated by market recovery.
> 0	<i>No Mitigation</i>	3	"Certain of the Company's component suppliers and logistical service providers experienced disruptions , resulting in supply shortages that affected sales worldwide."	Explicit link to worldwide sales impact (supply shortage).

Note. [†] The *Sample Score* column shows one representative component selected from the full Micro-Card assigned to the highlighted snippet.

Risk-Response-Driven Portfolio Strategy

Companies that proactively disclose risk responses are signaling transparency and superior risk management capabilities.

Hypothesis

Portfolios composed of companies with abundant (negative Response Scores) responses should systematically outperform portfolios with missing (positive/zero Response Scores) responses.

Portfolio Construction

Two equal-weighted portfolios of 20 S&P 500 stocks, rebalanced annually based on Response Score rank, avoiding look-ahead bias.

Economic Rationale

Disclosure signals management control and accountability, reducing information asymmetry and boosting investor confidence.

Empirical Results: Response Pays Off

Backtesting (July 2021 – June 2025) strongly validates the core hypothesis, showing significant performance divergence.

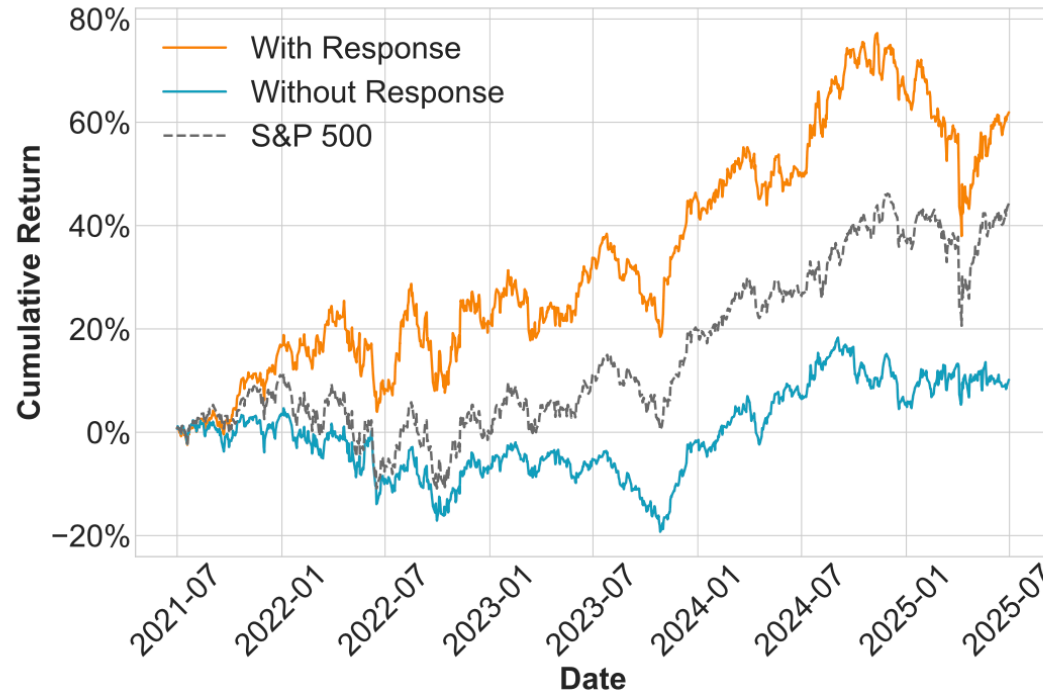


Figure 4: Portfolio Cumulative Returns, Action vs No Action (with baseline S&P 500)

13.78%

With Response
Annualized Return

3.75%

Without Response
Annualized Return

0.75

With Response
Sharp Ratio

0.23

Without Response
Sharp Ratio

*The "With Response" group outperformed the "Without Response" group by more than **three times** in annualized returns, demonstrating superior risk-adjusted performance.*

Conclusion: Bridging Disclosure and Asset Pricing



Novel Framework

DNA with CoT provides a verifiable method for deep semantic analysis of 10-K narratives.



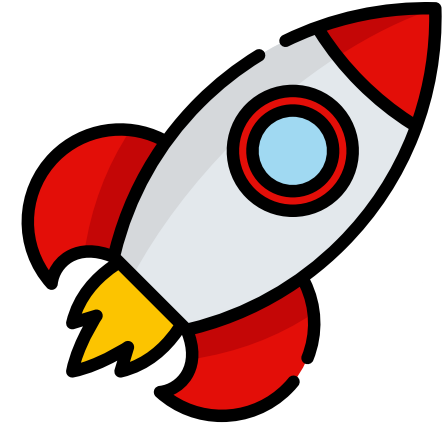
Dual Narrative

Successfully quantified the structural divergence between Item 1A (Risk) and Item 7 (Response).



Economic Value

Response Scores capture market inefficiency, generating significant excess returns for portfolio strategies.



Future Work

Response Scores capture market inefficiency, generating significant excess returns for portfolio strategies.

Thank you!

Q&A



"Two Sides of the Same Coin":

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